Exam Equations

Trigonometry: $\sin\Theta = h_o/h$ $\Theta = \sin^{-1}(h_o/h)$ $\cos\Theta = h_a/h$ $\Theta = \cos^{-1}(h_a/h)$ $\tan\Theta = h_o/h_a$ $\Theta = \tan^{-1}(h_o/h_a)$

Pythagorean Theorem $h^2 = h_o^2 + h_a^2$

$\cos(90° - \Theta) = \sin(\Theta)$ $\cos(90° + \Theta) = -\sin(\Theta)$ $\cos(180° - \Theta) = -\cos(\Theta)$
$\sin(90° - \Theta) = \cos(\Theta)$ $\sin(90° + \Theta) = \cos(\Theta)$ $\sin(180° - \Theta) = \sin(\Theta)$

Vectors: $\vec{\mathbf{a}} = (a_x)\hat{\mathbf{x}} + (a_y)\hat{\boldsymbol{y}}$ $\vec{\mathbf{a}} \bullet \hat{\boldsymbol{x}} = a_x = |\vec{a}|\cos(\alpha)$ $\vec{\mathbf{a}} \bullet \hat{\boldsymbol{y}} = a_y = |\vec{a}|\cos(\beta)$

$$\alpha = \cos^{-1}\left[\frac{a_x}{|\vec{\mathbf{a}}|}\right] \qquad \beta = \cos^{-1}\left[\frac{a_y}{|\vec{\mathbf{a}}|}\right] \qquad |\vec{\mathbf{a}}| = \sqrt{\vec{\mathbf{a}} \bullet \vec{\mathbf{a}}} = \sqrt{(a_x)^2 + (a_y)^2}$$

Dot product: $\vec{a} \bullet \vec{b} = |\vec{\mathbf{a}}||\vec{\mathbf{b}}|\cos(\varphi)$ Cross Product: $\vec{c} = \vec{a}x\vec{b} = \left[|\vec{\mathbf{a}}||\vec{\mathbf{b}}|\sin(\varphi)\right]\hat{c}$

Displacement: $\vec{\boldsymbol{x}} = \vec{r}_f - \vec{r}_i$ Speed = d/t velocity: $\vec{\mathbf{v}} = d\vec{\mathbf{x}}/dt$ acceleration: $\vec{a} = d\vec{\mathbf{v}}/dt$

Newton's Laws of Motion $\Sigma\vec{\mathbf{F}} = m\vec{\boldsymbol{a}}$ or $\vec{\boldsymbol{a}} = \Sigma\vec{\boldsymbol{F}}/m$ Weight: $\vec{W} = m\vec{\boldsymbol{g}}$, $|\vec{\boldsymbol{g}}| = 9.8$ N/kg

$\Sigma\vec{\mathbf{F}} = \vec{\mathbf{F}}_{\mathbf{net}} = (\Sigma F_x)\hat{\boldsymbol{x}} + (\Sigma F_y)\hat{\boldsymbol{y}}$ where $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$

For equilibrium: $\Sigma F_x = 0$ and $\Sigma F_y = 0$ For non-equilibrium: $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$

Centripetal Acceleration and Force, $a_c = v^2/r = r\omega^2$, $F_c = ma_c = m\,v^2/r$ Power: $P = dW/dt = dE/dt$ $P_{avg} = \Delta W/\Delta t$

Kinetic Energy: $KE = \frac{1}{2}\,mv^2$ Gravitational Potential Energy: $PE_g = mg(y_f - y_i) = mgh$

Elastic Potential Energy: $PE_e = \frac{1}{2}\,kx^2$, Springs: $F = k\Delta x$,, Conservation of Energy, $\Delta KE + \Delta PE = 0$

Work /Energy, $W = \Delta KE + \Delta PE = (F\cos\varphi)\Delta x$ Total work: $W_{net} = W_1 + W_2 + W_3 + \ldots$ or $W_{net} = (\Sigma F)x$

Current: $I = dq/dt$, Charge: $Q = Ne$, Ohm's Law, $V = IR$, Electrical Power, $P = IV = I^2R = V^2/R$, P = Energy/time

Peak and rms current/voltage: $I_{rms} = I_p/\sqrt{2}$ $V_{rms} = V_p/\sqrt{2}$

Force on a current in a magnetic field: $|\vec{F}_M| = ILB\sin\phi$, Torque on a Current Carrying Coil: $|\vec{\tau}| = NIAB\sin\varphi$

$\vec{\boldsymbol{B}}$ for Long straight Wire: $|\vec{\mathbf{B}}| = \mu_o I/2\pi r$, $\vec{\boldsymbol{B}}$ at Center of circular coil - $|\vec{\mathbf{B}}| = \mu_o IN/2r$

$\vec{\boldsymbol{B}}$ for long solenoid $|\vec{\mathbf{B}}| = \mu_o NI/L$ Net magnetic field: $\Sigma\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 + \cdots$

Magnetic Flux (Φ_M): $\Phi_M = B\cos\varphi A$, Faraday's Law: $V_{emf} = -N\,\Delta\Phi_M/\Delta t$, $\Delta\Phi = B_fA_f\cos\varphi_f - B_iA_i\cos\varphi_i$

$$\Phi_M = \int \vec{B} \bullet \hat{n}dA \qquad V_{emf} = -N\frac{d}{dt}\left[\int \vec{B} \bullet \hat{n}dA\right] \qquad \text{Uniform } \vec{B} \text{ field } V_{emf} = -N\frac{d\Phi}{dt}$$

Motional emf: $V_{emf} = vBL\sin(\phi)$ Generator peak : $V_p = NAB2\pi f$ Electric motor: $V_{applied} = V_{Backemf} + Ir$

Mutual Induction: V_{emf} (Secondary) $= -M(\Delta I_p/\Delta t)$, $M = N_s\Phi_s/I_p$, Transformers: $I_p/I_s = V_s/V_p = N_s/N_p$

Self-inductance: V_{emf} (Self) $= -L(\Delta I/\Delta t)$, $L = N\Phi/I$, $L(\text{solenoid}) = \mu N^2A/l$

Relationship between $\vec{\mathbf{E}}$ & $\vec{\mathbf{B}}$, $|\vec{\mathbf{E}}| = c|\vec{\mathbf{B}}|$, Energy in a photon: Energy = hf, $h = 6.63 \times 10^{-34}$ Js

Wave equation $v = f\lambda$ EM waves in Vacuum/air: $c = f\lambda$ Speed of light in vacuum $c = 3 \times 10^8$ m/s

Blackbody radiation: Peak $\lambda = (2.898 \times 10^{-3}\ \text{Km})/T$ where T is in Kelvin, $T_K = 273 + T_c$

Net Power $= e\sigma A(T^4 - T^4_o)$ $\sigma = 5.67 \times 10^{-8}\ \text{J/(sm}^2\text{K}^4\text{)}$ and e is the emissivity

EM wave Doppler shift $f' = f\sqrt{\frac{1+\frac{v}{c}}{1-\frac{v}{c}}}$

Intensity: $I = P/A$ $I = c\varepsilon_o E_{rms}^2$, $I = cB_{rms}^2/\mu_o$ Intensity of a Point source: $I = P/(4\pi r^2)$ Power: P = Energy/time

$c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C^2/Nm^2

Power at angle and efficiency η: $P = \eta[IA\cos(\varphi)]$ Efficiency η = useful energy/total energy

Un-polarized light through polarizer: $I = \frac{1}{2} I_o$ Polarized light through polarizer: $I = I_o(\cos\Theta)^2$

Law of Reflection: $\Theta_i = \Theta_r$

$k = 9.0 \times 10^9$ Nm^2/C^2, $\varepsilon_o = 8.85 \times 10^{-12}$ C^2/Nm^2, $\mu_o = 1.257 \times 10^{-6}$ Tm/A, $c = 3 \times 10^8$ m/s
Giga (G) = 10^9, Mega (M) = 10^6, milli (m) = 10^{-3}, 1 micron (μ) = 10^{-6}, 1 nano (n) = 10^{-9}, 1 pico (p) = 10^{-12}

Rulers for Derivatives

$$Power\ rule: \ \frac{d}{dx}[x^n] = nx^{n-1}$$

$$Addition\ or\ subtraction: \ \frac{d}{dx}[y \pm u] = \frac{d}{dx}[y] \pm \frac{d}{dx}[u] \qquad Product\ rule: \ \frac{d}{dx}[yu] = u\frac{d}{dx}[y] + y\frac{d}{dx}[u]$$

$$\frac{d}{dt}[\sin(\omega t)] = \omega\cos(\omega t) \qquad \frac{d}{dt}[\cos(\omega t)] = -\omega\sin(\omega t) \qquad \frac{d}{dt}[e^{at}] = ae^{at}$$

Integration Rules

$$\int (0)dx = C \qquad \int (1)dx = x + C \qquad \int (mx)dx = m\int (x)dx \qquad where\ C, a\ and\ m\ are\ constants$$

$$Addition\ and\ subtraction: \ \int (u+v)dx = \int (u)dx + \int (v)dx$$

$$Power\ Rule \quad \int x^n\,dx = \frac{x^{n+1}}{n+1} \qquad \int \sin(mx)\,dx = -\frac{\cos(mx)}{m} \qquad \int \cos(mx)\,dx = \frac{\sin(mx)}{m} \qquad \int e^{at}dt = \frac{1}{a}e^{at}$$

$$Definate\ Integral: \ \int_a^b [Integrand]dx = [Integral]\,|_a^b = \text{Integral with (b)} - \text{Intergral with (a)}$$

$$\int \frac{1}{x}dx = ln[x] \qquad \int \frac{1}{(x^2 \pm a^2)^{\frac{1}{2}}}dx = ln\left[x + \sqrt{x^2 \pm a^2}\right] \qquad \int \frac{x}{(x^2 \pm a^2)^{\frac{1}{2}}}dx = \sqrt{x^2 \pm a^2}$$

$$\int \frac{1}{(x^2+a^2)^{3/2}}dx = \frac{x}{a^2(x^2+a^2)^{1/2}} \qquad \int \frac{x}{(x^2+a^2)^{3/2}}dx = -\frac{1}{(x^2+a^2)^{1/2}}$$